Evaluation metrics

Token level (masked gap tokens $\mathcal{M}$)

Let K be the number of codebooks, $y_{k,t}$ the true code, and $p_\theta(\cdot \mid k,t)$ the predicted distribution.

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathcal{M}|} \sum_{(k,t)\in\mathcal{M}} \ln p_\theta(y_{k,t} \mid k,t) \qquad \text{[nats/token]} \tag{1}$$

$$\text{PPL} = \exp\big(\mathcal{L}_{\text{CE}}\big) \qquad \text{[unitless; chance = 1024]} \tag{2}$$

$$\text{acc}_{\text{top-}n} = \frac{1}{|\mathcal{M}|} \sum_{(k,t)\in\mathcal{M}} \mathbb{1}\big[y_{k,t} \in \text{top}_n\, p_\theta(\cdot \mid k,t)\big] \qquad \text{[fraction, 0–1]} \tag{3}$$

Audio level (decoded gap, $\hat{x}$ prediction, x target)

SI-SDR [dB, higher is better]

$$\alpha = \frac{\langle \hat{x}, x\rangle}{\langle x, x\rangle + \varepsilon}, \qquad \text{SI-SDR} = 10\log_{10}\frac{\|\alpha x\|^2}{\|\hat{x} - \alpha x\|^2} \tag{4}$$

Spectral convergence [dimensionless, lower is better]. With $S_r(\cdot) = |\text{STFT}_r(\cdot)|$ at resolution $r \in \{1,\dots,R\}$:

$$\text{SC} = \frac{1}{R}\sum_{r=1}^{R} \frac{\|S_r(x) - S_r(\hat{x})\|_F}{\|S_r(x)\|_F} \tag{5}$$

Log-magnitude loss [nats, lower is better]

$$\mathcal{L}_{\text{mag}} = \frac{1}{R}\sum_{r=1}^{R} \text{mean}\,\big|\ln S_r(x) - \ln S_r(\hat{x})\big| \tag{6}$$

Total multi-resolution STFT loss with weights $\lambda_{\text{sc}}, \lambda_{\text{mag}}$:

$$\mathcal{L}_{\text{STFT}} = \lambda_{\text{sc}}\,\text{SC} + \lambda_{\text{mag}}\,\mathcal{L}_{\text{mag}} \tag{7}$$

Waveform L1 [normalized amplitude, full scale = 1]

$$\mathcal{L}_{\text{L1}} = \frac{1}{N}\sum_{n=1}^{N} \big|x[n] - \hat{x}[n]\big| \tag{8}$$

SI-SDR and SC are computed on a *margin-included* crop (gap plus context frames) and a *gap-only* crop (margin = 0).

Units summary

Metric	Unit	Better
$\mathcal{L}_{\text{CE}}$	nats/token	lower
PPL	unitless	lower
acc top-1/5	fraction	higher
SI-SDR	dB	higher
SC	ratio	lower
$\mathcal{L}_{\text{mag}}$	nats	lower
$\mathcal{L}_{\text{L1}}$	normalized amplitude	lower